AI SITE HISTORY REPORT

502 Church Road, Taigum QLD 4018

Generated 07 Apr 2026 | Lat -27.333158, Lng 153.048732 | Confidence: Medium

Executive Summary

In addition to Former Pond A, a smaller secondary former pond feature lies on-site in the northwest corner, visible historically as a small circular water body with vegetated edges and a moisture signature in current imagery. A low-lying depression is also identified nearby. Off-site to the northeast, Cabbage Tree Creek presents a creek adjacency risk with moisture-retaining zones potentially affecting the site. Minor track formations and disturbed ground patches are evident in the current imagery, indicating localized fill or earthworks.

On-site: On-site features include the known Former Pond A and an additional smaller secondary former pond in the northwest corner, characterized by historical open water presence and persistent moisture signature. A low depression adjacent to this secondary pond shows a potential moisture-retaining area. Earthworks and track formations are present but limited in extent.

Adjacent / context: Adjacent to the northeast, Cabbage Tree Creek runs off-site providing natural drainage and increasing moisture-related risk near the property boundary. No adjacent off-site ponds or basins were detected within the wider context imagery.

Site Details

Address	502 Church Road, Taigum QLD 4018
Location source	polygon
Latitude / Longitude	-27.333158, 153.048732
Site bbox	122.8 m × 101.7 m
Context bbox	176.8 m × 155.7 m
Polygon supplied	Yes

Key Flags

Key Flag	Level	Comment
Former water features	High	Distinct former pond visible in multiple historical images dating back to 1995; strong circular morphology and vegetated ring.
Vegetation / land change	Low	Site shows some cleared areas around the former pond likely related to farmland or minor earthworks.
Fill / disturbance	Elevated	Disturbed ground patches adjacent to former pond indicate fill or earthworks.

Flag Summary

Potential Fill / Disturbance — Elevated

Disturbed ground patches adjacent to former pond indicate fill or earthworks.

Former Water Features — High

Distinct former pond visible in multiple historical images dating back to 1995; strong circular morphology and vegetated ring.

Vegetation / Land Change	**Low**
Site shows some cleared areas around the former pond likely related to farmland or minor earthworks.	

Screening Outcome

Additional smaller former pond and low depression features on-site present medium confidence moisture-related geotechnical risks. Adjacent creek adds drainage influence. Localized disturbance requires further field assessment.

Distinct Feature Review

Feature	Type	Relation	Confidence	Notes
Former Pond A	former_pond	on_site	high	Circular morphology with vegetated ring, open water clearly visible in 1995 and 2005 historical imagery, now mostly infilled with vegetation but still exhibiting moisture signature.
Disturbed Ground Patch A	disturbance	on_site	medium	Irregular patch of disturbed ground or fill adjacent to former pond with sparse vegetation, possibly associated with earthworks.
Secondary Former Pond B	former_pond	on_site	medium	Small circular feature with vegetated edge, visible as water in 1995 historical image and showing moisture presence currently.
Low-lying Depression C	depression	on_site	medium	Subtle topographic low adjacent to Secondary Former Pond B with darker vegetation indicative of moisture retention.
Track Formation D	track	on_site	medium	Dirt track or service road visible mostly in current subzone images connecting south side of site buildings to pond areas.
Cabbage Tree Creek	drainage_feature	adjacent	high	Major creek east and northeast of site boundary creating probable moisture-retaining geomorphic setting near site edges.

Feature Timeline

Former Pond A — 1995: Open water clearly visible with distinct circular pond morphology.

Former Pond A — 2005: Pond still visibly holding water with vegetated edges.

Former Pond A — 2011: Water presence reduced; pond partially infilled with more vegetation.

Former Pond A — 2021: Pond mostly infilled, showing dense vegetation and moist soil but no open standing water.

Secondary Former Pond B — 1995: Open water clearly visible with circular shape and vegetated margins.

Secondary Former Pond B — 2021: Mostly vegetated with moisture signature, water no longer open but morphology still present.

Low-lying Depression C — 1995: Slight darker vegetation within topographic low visible near Secondary Former Pond B.

Low-lying Depression C — 2021: Consistency of moisture signature maintained, indicating persistent wetness potential.

Track Formation D — 2021: Visible dirt track running through south side of site showing disturbance.

Detailed Findings

Visible Observations

- Circular former pond feature with vegetated berm visible across multiple imagery years
- Open water present in 1995 and 2005 imagery, infilled by 2011 and 2021
- Vegetated disturbed ground adjacent to former pond
- Nearby Cabbage Tree Creek to northeast off-site
- Secondary former pond feature on northwest site corner
- Low-lying depression adjacent to secondary pond
- Dirt access track crossing south of site building cluster
- Cabbage Tree Creek adjacent northeast boundary
- Vegetation patterns indicating moisture retention near ponds and depression

Overall Change Timeline

1995-2021: Former pond transitioned from open water body to primarily infilled vegetated depression, indicating possible historical drainage or sedimentation.

1995-2021: Main former pond and secondary pond features transitioned from open water to vegetated with moisture signatures; track and disturbance features emerged in recent years; creek adjacency remains constant.

Possible Geotechnical Implications

- possible soft or moisture-sensitive soils in former pond area
- moisture retention and drainage issues near former pond
- potential unstable ground in fill/disturbance zones
- possible soft or moisture-sensitive soils in former pond and depression areas
- drainage influences from adjacent creek creating seasonal moisture retention
- localized disturbance affecting soil strength near tracks

Targeted Investigation Focus

- target test pit or borehole in former pond area to assess soil moisture and fill
- inspect disturbed ground patch for fill origin and stability
- target test pit or borehole in secondary former pond area (Secondary Former Pond B)
- investigate low-lying depression moisture conditions
- verify fill extent and condition around track formation

Recommended Follow-Up

- field inspection to confirm former pond boundaries and soil conditions
- geotechnical subsurface investigation to assess moisture sensitivity
- field inspection focusing on soil moisture and disturbance extent
- topographic survey to confirm subtle depressions and pond boundaries

Supporting Imagery

current_wide_context — current

Source: mapbox_bbox_wide_context

current_context — current

Source: mapbox_bbox_context

current_site — current

Source: mapbox_bbox_site

current_subzone_west — current

Source: mapbox_bbox_subzone

current_subzone_east — current

Source: mapbox_bbox_subzone

historical_qld_followup_1 — 1995 (1995-06-18)

Source: qld_public_imagery | Queensland Digital Orthophoto Image Data - Brisbane_1995_60cm_9543 | Resolution: 60 2

historical_qld_followup_4 — 2015 (2015-05-30)

Source: qld_public_imagery | Queensland Digital Orthophoto Image Data - Brisbane_2015_10cm_SISP_LGA | Resolution: 10 2

historical_qld_followup_6 — 2021 (2021-07-17)

Source: qld_public_imagery | Queensland Digital Orthophoto Image Data - Brisbane_Urban_2021_10cm_SISP | Resolution: 10 2

Limitations & Disclaimer

- No recent ground truth data available to confirm current soil conditions
- Imagery resolution limits detailed mapping of small-scale fill or disturbance
- Temporal gaps in historical imagery between 2011 and 2021
- Limited resolution in historical imagery may affect detection certainty of minor features
- Surface moisture signatures inferred from vegetation only, no subsurface data
- Current imagery dates lack seasonal wetlands assessment

This AI site history scan is a preliminary screening tool only. It is intended to identify possible visual indicators for further professional review. It must not be relied on as a substitute for geotechnical investigation, engineering assessment, or design advice.